The OS provides access to block-level storage via open, write, and read system calls. The simplified flow of a read request goes like this:

1. An application wants to read the file /path/to/file.txt and makes a read system call.
2. The OS forwards the read request to the filesystem.
3. The filesystem translates /path/to/file.txt to the block on the disk where the data is stored.

Applications like databases that read or write files by using system calls must have access to block-level storage for persistence. You can't tell a MySQL database to store its files in an object store because MySQL uses system calls to access files.

Not all examples are covered by the Free Tier

The examples in this chapter are not all covered by the Free Tier. A warning message appears when an example incurs costs. Nevertheless, as long as you don't run all other examples longer than a few days, you won't pay anything for them. Keep in mind that this applies only if you created a fresh AWS account for this book and nothing else is going on in your AWS account. Try to complete the chapter within a few days; you'll clean up your account at the end.

AWS provides two kinds of block-level storage: